



PSG INSTITUTE OF TECHNOLOGY AND APPLIED RESEARCH

DEPARTMENT OF COMPUTER SCIENCE & ENGINEERING

SECOND PROJECT REVIEW

Project Title : **An XAI-Based Decision Support Framework for Semiconductor Fault Diagnosis: Validation Across Synthetic, Manufacturing, and Device-Aging Data**

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Outline of the Presentation

- Problem Statement
- Literature Review
- Our Approach
- Architecture Diagram and Flowchart
- Algorithms and Techniques Used
- Modules, Roles, and Responsibilities
- Snapshots and Results
- Future Enhancements
- References



PROBLEM STATEMENT

ML models for semiconductor fault diagnosis often report near-perfect accuracy (95–99%+), but this breaks down under real scrutiny — due to three key issues:

- **Data Leakage** - Preprocessing done before the train/test split lets test data "leak" into training, inflating results.
- **Misleading Accuracy** - With severe class imbalance (e.g., 14:1 in real fab data), accuracy hides poor fault-detection performance.
- **Unstable Explanations** - SHAP/LIME explanations are rarely checked for consistency across folds, making them unreliable for real decisions.

Literature Review

1. An Experimental Evaluation of Fault Diagnosis from Imbalanced and Incomplete Data for Smart Semiconductor Manufacturing

Authors: Salem Salem, Soheil Taheri, Jafar A. Alzubi, Xindong Wu, Shengrui Wang, Jiuyong Li, Jianjun Hu, Jian Cao & Yanchun Zhang (2018)

Venue: *Big Data and Cognitive Computing*

Core Idea: Systematic evaluation of 288 combinations of imputation, feature selection, resampling, and classifiers on the SECOM dataset. Introduces In-painting KNN Imputation for handling missing sensor data.

Research Gap: No leakage-safe (fold-confined) preprocessing or resampling protocol; feature importance derived from a single trained model without assessing explanation stability.

2. An Expandable Yield Prediction Framework Using Explainable AI for Semiconductor Manufacturing

Authors: Jonghyeok Lee & Hyunseok Roh (2023)

Venue: *Applied Sciences*

Core Idea: Integrates heterogeneous semiconductor manufacturing data for yield prediction, benchmarks multiple ML models, and interprets the best-performing Random Forest model using SHAP for actionable engineering insights.

Research Gap: SHAP explanations are presented without evaluating stability across folds or datasets; calibration and explanation reliability are not quantitatively assessed.

Literature Review

3. Fault Detection in Semiconductor Manufacturing Using Advanced Machine Learning Techniques with Interpretability

Authors: Swaroop Srisailam, Pallab Mandal, Kalur Lalith Aditya, Praveen Pankajakshan, et al. (2025)

Venue: *Springer Book Chapter*

Core Idea: Uses SMOTE/ADASYN, dimensionality reduction (590 → 33 features), and a voting ensemble (RF, SVM, KNN, LR, GB, XGB) on the SECOM dataset; reports 99.68% accuracy. **Research Gap:** Reported performance suggests possible protocol differences (e.g., oversampling before train/test separation); relies on accuracy as the headline metric despite severe class imbalance and does not report leakage-safe evaluation.

4. Shapley-based Explainable AI for Clustering Applications in Fault Diagnosis and Prognosis

Authors: Joseph Cohen, Xun Huan & Jun Ni (2024)

Venue: *Journal of Intelligent Manufacturing*

Core Idea: Extends SHAP from prediction to semi-supervised clustering for fault diagnosis and prognosis. Produces interpretable fault clusters and validates explanations using rule extraction on semiconductor and PHM datasets.

Research Gap: Focuses on clustering rather than supervised fault classification; explanation stability across folds, calibration, and robustness under class imbalance are not evaluated.

Literature Review

5. Rare Class Prediction Model for Smart Industry in Semiconductor Manufacturing

Authors: Abdelrahman Farrag, Mohammed-Khalil Ghali & Yu Jin (2024)

Venue: *arXiv*

Core Idea: Develops a rare-defect prediction framework using mixed imputation, feature voting, SMOTE, and XGBoost to improve minority-class detection in semiconductor manufacturing. **Research Gap:** No explainability (SHAP/LIME), no decision-support layer, no calibration or threshold optimization, and no explanation-stability analysis.

6. Explainable AutoML (xAutoML) with Adaptive Modeling for Yield Enhancement in Semiconductor Smart Manufacturing

Authors: Weihong Zhai, Xiupeng Shi, Yiik Diew Wong, Qing Han & Lisheng Chen (2024)

Venue: *arXiv*

Core Idea: Proposes an Explainable AutoML (xAutoML) framework that automates feature engineering, model selection, and hyperparameter optimization while incorporating explainability for semiconductor yield prediction and defect diagnosis. **Research Gap:** Focuses on AutoML optimization rather than rigorous evaluation methodology; lacks leakage-controlled benchmarking, statistical significance testing, explanation-stability analysis, robustness evaluation, and validation across multiple datasets/regimes.



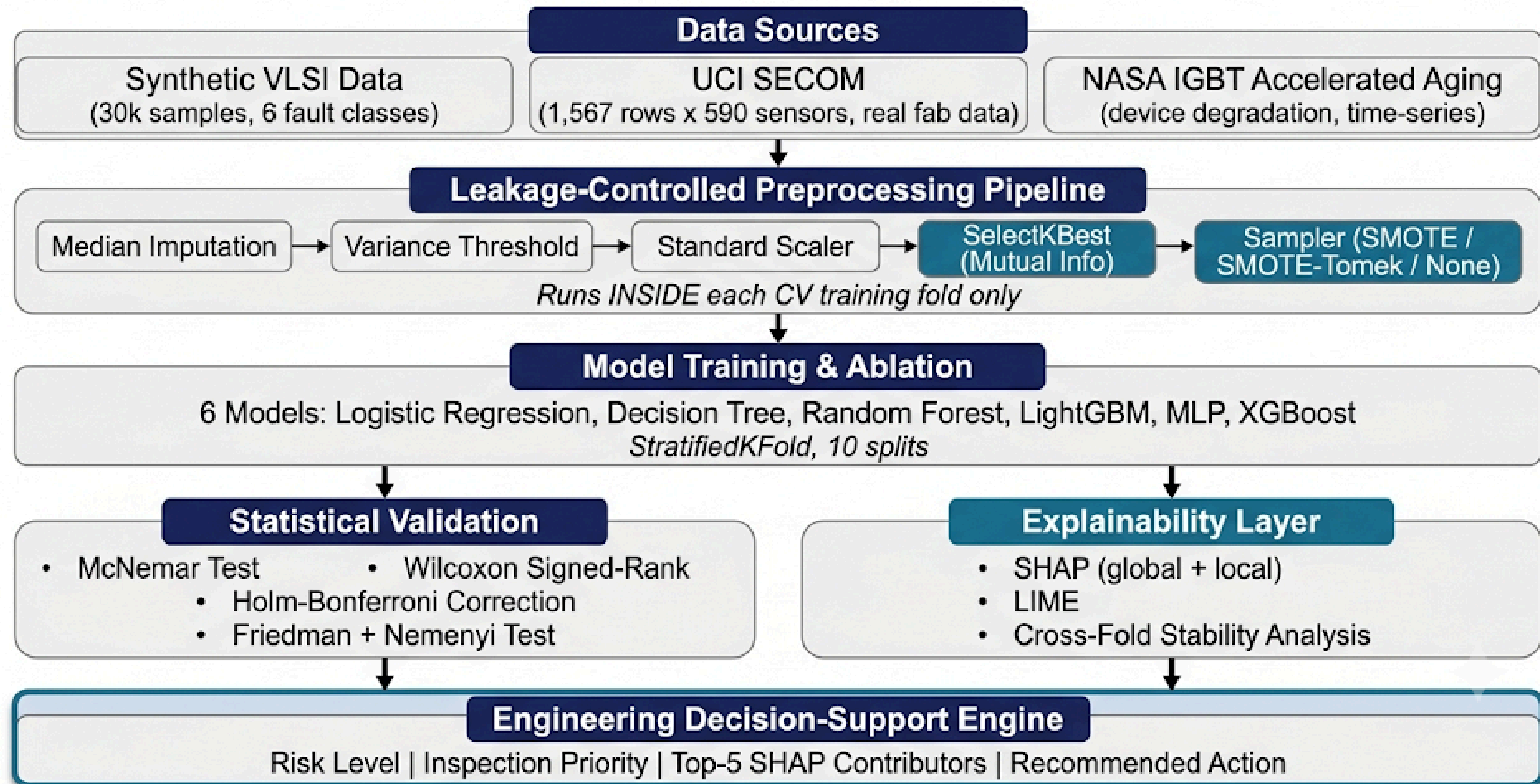
Our Approach

Not a new algorithm a validated decision-support framework for semiconductor fault diagnosis.

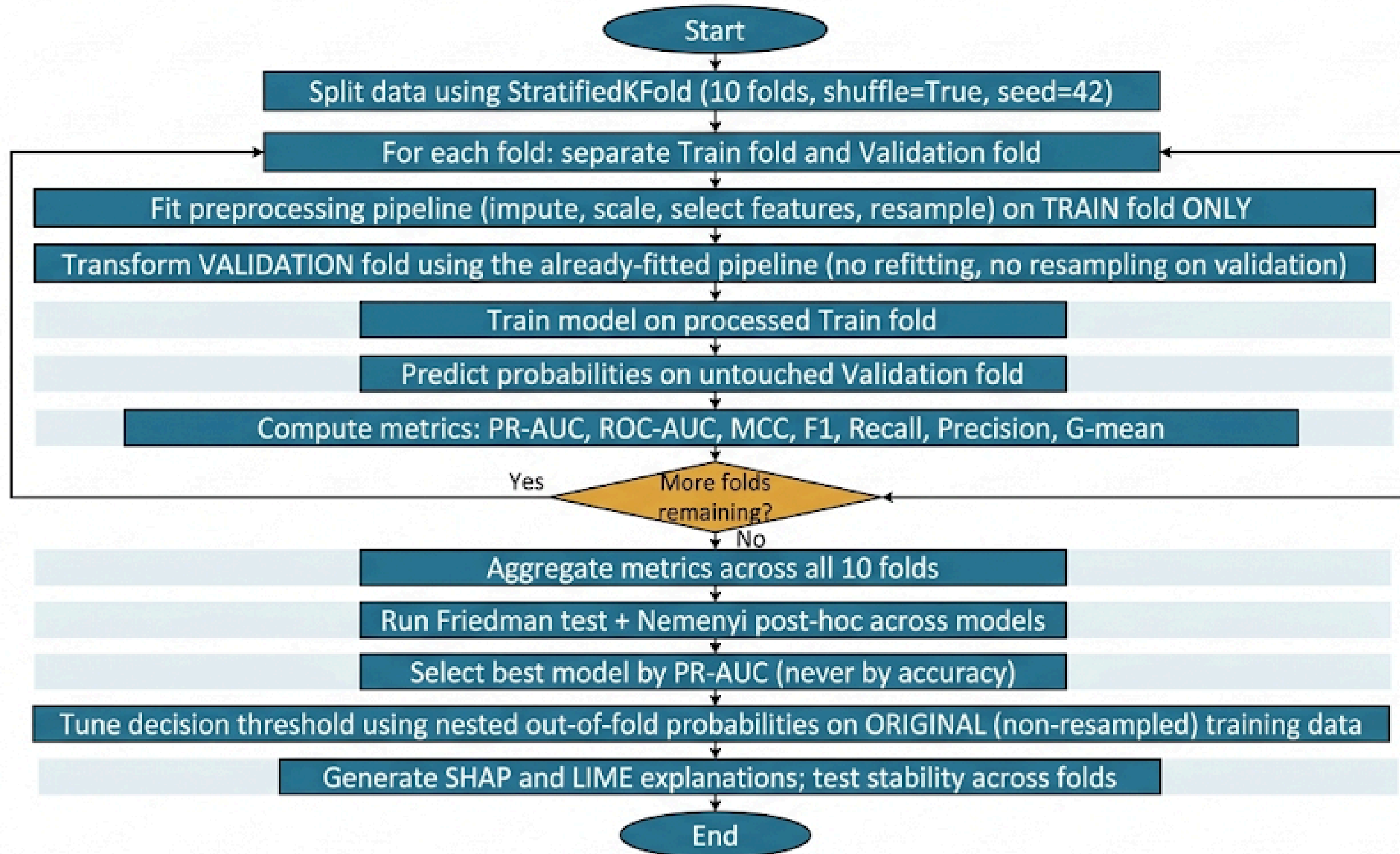
- **Leakage-Controlled Pipeline** - Preprocessing fit only inside each CV training fold, never on validation data.
- **Statistically Valid Comparison** - Six models compared using McNemar, Wilcoxon, and Friedman + Nemenyi tests, not raw accuracy.
- **Stable Explainability** - SHAP/LIME explanations tested for consistency across folds.
- **Robustness Testing** - Performance checked under noise, missing values, and data drift.
- **Decision Support Output** - Risk level, inspection priority, top contributors, recommended action.
- **Validated Across 3 Data Regimes** - Synthetic → real manufacturing (SECOM) → real device aging (NASA IGBT).

Architecture Diagram:

System Architecture — Explainable AI Framework for Semiconductor Fault Diagnosis



Flow Chart





Algorithm

Input: Dataset D, Models M, Imbalance strategies S, Feature counts K

Output: Best model, tuned threshold, validated explanations

- for each (model, strategy, k) in $M \times S \times K$:
- for each fold in StratifiedKFold(D, splits=10):
- Fit pipeline (impute $\rightarrow$ scale $\rightarrow$ select k $\rightarrow$ resample) on TRAIN fold only
- Transform VALIDATION fold (no refitting)
- Train model; predict on validation fold
- Record PR-AUC, ROC-AUC, MCC, F1, Recall
- Store mean $\pm$ std across folds
- Select best config = $\text{argmax}(\text{PR-AUC})$
- Validate significance: Friedman test + Nemenyi post-hoc
- Tune threshold using nested out-of-fold probabilities on ORIGINAL (non-resampled) training data
- Compute SHAP + LIME explanations for best model
- Test explanation stability across folds
- return best model, tuned threshold, validated explanations

Techniques Used for Implementation

- **Preprocessing:** median imputation, variance thresholding, standard scaling, mutual-information feature selection
- **Imbalance handling:** class weighting, SMOTE, SMOTE-Tomek (via imblearn.pipeline.Pipeline to prevent leakage)
- **Models:** Logistic Regression, Decision Tree, Random Forest, LightGBM, MLP, XGBoost
- **Validation:** StratifiedKFold (10-fold), RepeatedStratifiedKFold with 95% CIs, McNemar's test, Wilcoxon signed-rank test, Holm–Bonferroni correction, Friedman test + Nemenyi post hoc
- **Explainability:** SHAP (beeswarm, bar, waterfall, dependence plots), LIME, gain-vs-SHAP-vs-permutation agreement (Spearman), cross-fold SHAP stability (top-20 overlap + mean pairwise Spearman)
- **Robustness testing:** Gaussian noise injection (2/5/10/15%) on test data only, missing-value masking (5/10/20%), chronological drift split, feature ablation with random-control baseline



Datset Specification:

Attribute	Dataset 1 — Synthetic VLSI	Dataset 2 — UCI SECOM (Real Fab)	Dataset 3 — NASA IGBT Aging
Source	Physics-parameterized generator (released)	UCI ML Repository	NASA PCoE Prognostics Repository
Samples	30,000	1,567	6 devices (time-series)
Features	18 electrical/thermal parameters	590 anonymised process sensors	VCE(on), collector current, case temperature → window statistics
Target	6 fault classes (No Fault, Thermal, Timing, Power, Signal Integrity, Reliability)	Binary Pass/Fail	Degradation state (thermal overstress to latch-up)
Role	Framework demonstration & explanation-faithfulness validation	Real-world manufacturing validation	Device-reliability validation



Software Specification

Software Tool	Specification
ML Frameworks	scikit-learn, XGBoost, LightGBM
Imbalanced Learning	imbalanced-learn (SMOTE, SMOTETomek, leakage-safe Pipeline)
Explainability	SHAP (TreeExplainer), LIME
Statistics	SciPy (McNemar, Wilcoxon, Friedman), statsmodels
Data Handling / Plots	pandas, NumPy, Matplotlib, Seaborn (300 DPI exports)

Module Breakdown

Phase	Module	Description	Status
1	Data Acquisition & Preprocessing	Load datasets, analyze missing values and imbalance, remove unsuitable features, and perform leakage-safe preprocessing within CV folds.	✅ Completed
2	Model Benchmarking & Selection	Compare six ML models using multiple sampling and feature-selection strategies with repeated stratified 10-fold CV; select the best model.	✅ Completed – XGBoost selected (PR-AUC: 0.209, ROC-AUC: 0.725, Recall: 0.493)
3	Statistical Validation & Optimization	Perform statistical significance tests, hyperparameter tuning, confidence intervals, and probability calibration.	🔄 In Progress
4	Explainable AI & Robustness	Generate SHAP and LIME explanations and evaluate explanation stability and robustness.	⌚ Pending
5	Decision Support & Reporting	Produce risk predictions, recommended actions, publication-ready figures, and reproducible code release.	⌚ Pending



Gantt Chart

Task	Week 1	Week 2	Week 3	Week 4	Week 5	Week 6	Week 7
Fix McNemar, ECE & SHAP Stability	✓						
Ablation Study, Robustness & Drift Split		✓					
NASA IGBT Dataset Pipeline			✓	✓			
SHAP Faithfulness (Synthetic Dataset)			✓				
Decision-Support Engine & Export Module				✓			
Methodology & Data Leakage Section	✓						
Related Work (40+ Verified References)		✓	✓				
Results Section (Three Experimental Regimes)				✓	✓		
Discussion & Limitations					✓	✓	
Introduction, Abstract & Final Revision						✓	✓
GitHub Repository & Reproducibility Package					✓		
Publication Figures & Tables						✓	
Internal Review & Journal Submission							✓

Member	Role	Key Responsibilities
Adhithya J	Experiments	Experimental pipeline, Phase 5–9 fixes, ablation & robustness studies, NASA IGBT adaptation, decision-support engine, final results.
Arya A	Writing & Literature	Related Work (40+ verified references), Introduction, reference management, journal formatting.
Abishek N	Validation & Statistics	Verify data leakage, validate statistical tests (McNemar/Wilcoxon/Friedman), calibration metrics, reproduce results.
AshokKumar A	Figures & Reproducibility	Publication figures, GitHub repository, documentation, presentation slides, submission checklist & cover letter.

Future Enhancements:

- **Third data regime:** Validate the framework on NASA PCoE IGBT/MOSFET accelerated thermal-aging data, extending from yield prediction to device-reliability prognosis.
- **Explanation faithfulness:** Quantitatively verify SHAP rankings against the known ground-truth rules of the synthetic generator — a validation real data cannot provide.
- **Temporal deployment:** Extend the chronological-split drift analysis toward online monitoring with concept-drift detection and periodic recalibration.
- **Validated recommendations:** Replace the illustrative sensor-to-action mapping with expert-labelled mitigation actions from process engineers.
- **Cost-sensitive thresholds:** Set decision thresholds from actual misclassification costs (escaped defect vs. unnecessary inspection) rather than F1.
- **Broader data:** Extend to multi-fab / multi-product datasets and explore federated learning for cross-site diagnosis without sharing proprietary sensor data.

- 1.** M. Salem, S. Taheri, and J.-S. Yuan, "An Experimental Evaluation of Fault Diagnosis from Imbalanced and Incomplete Data for Smart Semiconductor Manufacturing," *Big Data and Cognitive Computing*, vol. 2, no. 4, p. 30, 2018.
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- 3.** S. Srisailam, P. Mandal, K. L. Aditya, P. Pankajakshan, and K. T. Rajamani, "Fault Detection in Semiconductor Manufacturing Using Advanced Machine Learning Techniques with Interpretability," in *Advanced Network Technologies and Intelligent Computing (ANTIC 2024)*, *Communications in Computer and Information Science*, vol. 2336, Springer, Cham, 2025.
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- 4.** J. Cohen, X. Huan, and J. Ni, "Shapley-based Explainable AI for Clustering Applications in Fault Diagnosis and Prognosis," *arXiv*, 2023.
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- 5.** A. Farrag, M.-K. Ghali, and Y. Jin, "Rare Class Prediction Model for Smart Industry in Semiconductor Manufacturing," *arXiv*, 2024.
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